26900042 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCBIF300 - PERL & BIOPERL PROGRAMMING

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **5** questions

Each question carries **2** marks

1. Define an array element in Perl. [K] / [CO1]
2. Mention the operator used to read data from a file. [K] / [CO1]
3. State the year in which Perl was originally developed. [K] / [CO1]
4. What is meant by finding all occurrences of a pattern in Perl? [K] / [CO3]
5. Show the general syntax of a Perl built-in function. [K] / [CO4]
6. Apply BioPerl to retrieve a GenBank record and obtain its sequence identifier. [A] / [CO6]
7. Apply the appropriate BioPerl module to create a simple BioPerl program. [A] / [CO6]

[2x5 = 10]

Part B

Short Essay

Answer any **4** questions

Each question carries **5** marks

8. Explain the purpose of the .pl file extension in Perl programming [K] / [CO1]

9. Describe the working of the for loop and foreach loop in Perl. [U] / [CO2]

10. Explain the purpose of the while loop in Perl. [U] / [CO2]

11. Describe how wildcard-like patterns can be represented using regular expressions in Perl. [K] / [CO3]

12. Generate a suitable method for designing subroutines that accept arguments and return values in Perl programming. [C] / [CO5]

13. Use `Bio::AlignIO` to read an alignment and obtain the sequence information from the alignment. [A] / [CO6]

[5x4 = 20]

Part C

Long Essay

Answer any **2** questions

Each question carries **10** marks

14. Explain operator precedence and associativity in Perl with suitable examples. [K] / [CO1]

15. Explain the procedure for executing a Perl script from the command line with suitable examples. [U] / [CO1]

16. Explain how character sets can be used for pattern matching in Perl. Illustrate your answer with different examples. [K] / [CO3]

17. Identify the different types of anchors used in Perl pattern matching and state the purpose of each. [K] / [CO3]

[10x2 = 20]